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Mid Term - Exploratory Data Analysis

ITAI-1371

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Patient Health Records Dataset Proposal

Dataset Overview

Our group chose the Patient Health Records dataset from Kaggle because it gives us a realistic data-cleaning challenge instead of data that is already prepared for machine learning. The raw file contains 10,000 patient records and 17 columns with demographic, health, lifestyle, medication, visit, and disease information. It includes both numerical and categorical data, along with missing values, inconsistent labels, mixed formats, and incorrect entries. Examples include different spellings for the same city, several forms of Yes/No values, blood pressure stored in different formats, and numerical columns that contain text or unrealistic values. Moreover, the target variable, Has_Disease, makes the dataset suitable for a binary classification problem.

Project Proposal

Our goal is to turn the raw dataset into a clean, numerical dataset that we can use for machine learning by working in Python through Jupyter Notebook. The process starts by preparing the target labels and splitting the data into 70% training and 30% testing. EDA will focus only on the training set, while the testing set stays separate during cleaning and preprocessing. This will help us avoid using information from the test set when making decisions about how to prepare the training data.

During preprocessing, the main focus will be on correcting data-quality problems and preparing the training data for machine learning. Missing values, invalid numerical entries, inconsistent categories, mixed formats, and unnecessary identifying information will need appropriate handling. Numerical and categorical features will be converted into consistent forms that a machine learning model can use, while useful information from the existing health data can be preserved or reorganized when needed. The target classes will also be checked for imbalance, and important preprocessing changes will be reviewed through printed samples and visual comparisons. Every change will come from Python code instead of manual edits to the dataset. The goal is to keep useful health information while removing or transforming values that could confuse a future machine learning model.

Matplotlib will show important changes before and after preprocessing. Printed samples will make each step easier to follow. At the end of the project, we expect a clean numerical training dataset with no missing feature values or text-based predictors. The processed dataset can then support a machine learning model that predicts Has_Disease.